

Kunal Mishra, Ph.D.

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 [kunal-mishra](#) |  [Google Scholar](#) |  [freeribosomes](#)

PROFILE

Computational biologist working on target discovery and translational biology in immune-mediated and fibrotic disease. I build pipelines that integrate human genetics, bulk, single-cell and spatial transcriptomics, and metabolomics to nominate causal genes, then take those candidates to the bench myself for functional validation. Recent work spans a cross-species systems genetics framework for causal gene discovery in kidney disease, conserved pro-fibrotic macrophage states across organs and species, and interferon-driven macrophage biology in lupus and other autoimmune settings.

SKILLS

- **Human Genetics and Systems Genetics:** GWAS interpretation and integration, eQTL and QTL mapping, rare variant burden testing (SKAT-O), whole genome sequencing analysis, co-expression network construction, causal gene prioritisation, cross-species evidence integration
- **Omics and Data Analysis:** Bulk RNA-seq (DESeq2, GSEA), single-cell and single-nuclei RNA-seq (Seurat, Scanpy, CCA/RPCA integration, SCTransform), spatial transcriptomics, cell-cell communication analysis (CellChat), untargeted and targeted LC-MS metabolomics, spectral flow cytometry (Cytek Aurora, CytoExploreR)
- **Engineering and Platforms:** R, Python, Bash, SQL; Snakemake, Git, conda; reproducible and standardised workflow development; HPC (PBS) and cloud research platforms (All of Us Researcher Workbench, BigQuery, Google Cloud Storage); public resources including GEO, ArrayExpress, Open Targets, Human Cell Atlas
- **Experimental:** Primary human monocyte-derived macrophage culture and differentiation, cell line, flow cytometry, siRNA knockdown, qPCR, ELISA, IF/IHC, bulk RNA-seq library preparation, mouse and rat models

EDUCATION

- **Doctor of Philosophy (Ph.D.) in Computational Biology** Aug 2021 - Nov 2025
Centre for Computational Biology, Duke-NUS Medical School Singapore
 - Thesis: multi-omic definition of the role of macrophages in chronic kidney disease. Performed all computational analyses and end-to-end experimental validation
- **Bachelor of Science (BSc. Hons.) in Biological Sciences and Medicinal Chemistry** Aug 2017 - Jul 2021
Nanyang Technological University Singapore
 - Graduated with Distinction; cumulative GPA 4.42 / 5.00
 - Relevant coursework: Bioinformatics Methods I & II, Metabolomics and Lipidomics, Advanced Immunology, Molecular & Cell Biology I & II, Organic Chemistry I-III, Drug Discovery and Development

RESEARCH EXPERIENCE

- **Computational Research Fellow** Dec 2025 - Present
Behmoaras Lab, Centre for Computational Biology, Duke-NUS Medical School Singapore
 - Lead kidney-focused research projects integrating systems genetics, multi-omics analysis, and experimental validation to identify causal genes and pathways in chronic kidney disease
 - Develop and apply reproducible computational pipelines for bulk, single-cell, and spatial transcriptomic analyses
 - Communicate findings as first-author and co-author manuscripts and through internal and external presentations
- **Bioinformatics Research Intern** May 2020 - Aug 2021
*Prabhakar Lab, Genome Institute of Singapore, A*STAR* Singapore
 - Developed and improved the performance of a novel single-cell feature selection algorithm (DUBStepR, *Nature Communications*, 2021)
 - Analysed single-nuclei transcriptomes from human fetal brains (Simons Foundation Autism Research Initiative-funded project)
 - Analysed pilot rheumatoid arthritis single-cell RNA sequencing data

PUBLICATIONS

AUTHOR NAME UNDERLINED; ‡ DENOTES EQUAL CONTRIBUTION

- [1] Mishra K, Sakban RB, Shankar S, Wong J, Farah BL, Guo J, Ching J, Tham MS, Kovalik JP, Gurley SB, Petretto E, Tolwinski NS, Coffman TM, Behmoaras J. **A Cross-Species Systems Genetics Framework Identifies Causal Genes in Diabetic Nephropathy** (2026). *bioRxiv* [preprint]. doi:10.64898/2026.06.30.735464. Under review.
- [2] Chen T[‡], Quinn Z[‡], Mishra K[‡], O'Connor EC[‡], Silver SE, Zhang Y, Valencia MJ, Mei Y, Behmoaras J, Ferreira LMR, Chatterjee P. **moPPIt: De novo generation of motif-specific and functionally active peptide binders via discrete flow matching** (2026). *bioRxiv* [preprint]. doi:10.1101/2024.07.31.606098. Under review.
- [3] Wong BH, Mishra K, Chin CF, Galam DLA, Tan BC, Behmoaras J, Chua AWC, Silver DL. **Mfsd2a is important for maintaining epidermal homeostasis** (2026). *PNAS*, 123(8). DOI: 10.1073/pnas.2531159123.
- [4] Yen YH, Chew G, Chong CM, Nithianandam V, Kang PJ, Wee H, Yuan F, Mishra K, Behmoaras J, Cheung C, Zeng L, Su H, Ng ASL, Feany MB, Zhang SC. **FLT1-ICD loss orchestrates endothelial senescence and vascular dysfunction in Alzheimer's disease** (2025). Under review.
- [5] Ouyang JF[‡], Mishra K[‡], Xie Y, Park H, Huang KY, Petretto E, Behmoaras J. **Systems-level identification of a matrisome-associated macrophage polarisation state in multi-organ fibrosis** (2023). *eLife*, 12:e85530. DOI: 10.7554/eLife.85530
- [6] Ranjan B, Sun W, Park J, Mishra K, Schmidt F, Xie R, Alipour F, Singhal V, Joanito I, Honardoost MA, Yong JMY, Koh ET, Leong KP, Rayan NA, Lim MGL, Prabhakar S. **DUBStepR is a scalable correlation-based feature selection method for accurately clustering single-cell data** (2021). *Nature Communications*, 12:5849. DOI: 10.1038/s41467-021-26085-2

CONFERENCE PRESENTATIONS

Pro-fibrotic SPP1⁺ macrophages in disease

- **European Alliance of Associations for Rheumatology (EULAR) Symposium** 2026
Poster Tour. London, United Kingdom
- **6th CSI-JSI-KAI-SgSI Joint Symposium** 2026
Poster. Seoul, South Korea
- **Human Cell Atlas (General Meeting)** 2025
Poster. Singapore
- **2nd Spatial Biology Congress: Asia** 2024
Poster. Singapore

Systems genetics for diabetic nephropathy

- **American Society of Nephrology (Kidney Week)** 2025
Poster. Houston, USA
- **Advances in Cardio-Renal-Metabolic Diseases (Duke-NUS)** 2024
Oral Talk. Singapore

Re-purposing anti-cancer compounds as anti-malarials

- **International Conference of Undergraduate Research** 2019
Oral Talk. Virtual
- **Australasian Conference for Undergraduate Research** 2019
Oral Talk. Newcastle, Australia

RESEARCH GRANTS

- **Centre for Technology and Development Grant (SGD 50,000)** 2026
Duke-NUS Medical School, Singapore
- **NISTH-Micron Responsible AI Grant (SGD 5,000)** 2019
Nanyang Technological University, Singapore

HONOURS AND AWARDS

- **EULAR Travel Bursary Award** 2026
European Alliance of Associations for Rheumatology, Switzerland
- **Duke-NUS Achievement Prize, Outstanding Ph.D. Student** 2026
Duke-NUS Medical School, Singapore
- **Best Lightning Talk, Advances in Cardio-Renal-Metabolic Diseases** 2024
Duke-NUS Medical School, Singapore

- **Best Poster Award, Ph.D. Student Symposium** 2023
Duke-NUS Medical School, Singapore
- **Khoo Pre-Doctoral Fellowship** 2021
Duke-NUS Medical School, Singapore
- **Best Oral Presentation, Australasian Conference of Undergraduate Research** 2019
The University of Newcastle, Australia
- **Presidential Research Scholar** 2019
Nanyang Technological University, Singapore
- **Mr. and Mrs. Kwok Chin Yan Award for Student Initiative** 2019
Nanyang Technological University, Singapore

TEACHING AND MENTORSHIP

- **Peer Mentor** 2026 - Present
Duke-NUS Medical School, Singapore
 - Mentor incoming Ph.D. students on coursework, lab rotations, and thesis planning

SERVICE AND OUTREACH

- **Committee Member** 2026 - Present
Graduate Studies Curriculum Committee, Duke-NUS Medical School, Singapore
- **Panelist** 2024
International Women in STEMM Symposium, Singapore
- **Student Volunteer** 2022 - 2025
Duke-NUS Student Affairs, Singapore

MEDIA COVERAGE

- **Duke-NUS' Class of 2026 begin medicine's next chapter with compassion and purpose** 2026
Duke-NUS NewsHub 
- **SGH to lead \$5.8m research project to combat deadly autoimmune disease** 2024
The Straits Times; Singapore General Hospital (SGH)  
- **Scanning scars, Duke-NUS scientists find a 'scarcode' common across organs** 2023
Duke-NUS NewsHub 

ADDITIONAL INFORMATION

Languages: English (Proficient), Hindi (Proficient), Indonesian (Conversational), French (Elementary)

Interests: Music (Guitar, Classical Flute), Photography